Social Media Platform Database Design and Analysis using SQL

Objective

- The objective of this project is to design and implement a relational database system that simulates a social media platform.
- The project focuses on storing user data, managing posts, tracking interactions (likes, comments, followers), and generating analytical insights from the data.

Database Description

The database represents a simplified social media application where users can:

- Create posts
- Like and comment on posts
- Follow other users
- Use hashtags

Tables and Structure

User_Profiles

Column Name	Data Type	Description
User_ID	INT (PK)	Unique user identifier
User_Name	VARCHAR(50)	Name of the user
Date_of_Birth	DATE	User's date of birth
Gender	ENUM	Gender (Male/Female)
Mail_ID	VARCHAR(20)	Unique email address

Post_Records

Column Name	Data Type	Description
Post_ID	INT (PK)	Unique post ID
User_ID	INT (FK)	References User_Profiles
Caption	TEXT	Post content
Posted_at	TIMESTAMP	Post creation time

Followers

Column Name	Description
Follower_ID	User who follows
Following_ID	User being followed
Follow_date	Timestamp of follow

Comments

Column Name	Description
Comment_ID	Primary key
Post_ID	Related post
User_ID	Comment author
Comment_text	Comment content
Commented_at	Timestamp

Likes

Column Name	Description
Like_ID	Primary key
Post_ID	Related post
User_ID	User who liked
Liked_at	Timestamp

Hashtags

Column Name	Description
Post_ID	Related post
Hashtag	Hashtag text

Notifications

Column Name	Description
Notification_ID	Primary key
User_ID	User receiving notification
Message	Notification text
Created_at	Timestamp

Features Implemented

The project includes the following SQL features:

- Data Definition Language (DDL)
- Data Manipulation Language (DML)
- Joins (Inner Join, Left Join)
- Aggregate functions (COUNT, AVG)
- Subqueries
- Stored procedures
- User-defined functions
- Views
- Triggers

Key Insights

- A small number of users generate higher engagement compared to others. These users can be identified as influencers.
- Only a subset of posts receives high interaction, indicating that content performance varies significantly.
- User posting behavior is not consistent across all dates. Activity tends to vary, indicating peak and low usage periods.
- Certain users show recent growth in followers, suggesting emerging influencers.
- Some hashtags appear frequently, indicating popular topics and trends among users.
- Engagement is unevenly distributed across posts, with some posts receiving significantly more likes and comments.

Conclusion

- This project demonstrates the design and implementation of a relational database system for a social media platform.
- It covers core SQL concepts and applies them to real-world scenarios such as engagement tracking, trend analysis, and user behavior analysis.
- The system provides meaningful insights that can be used for decision-making and improving user experience.

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